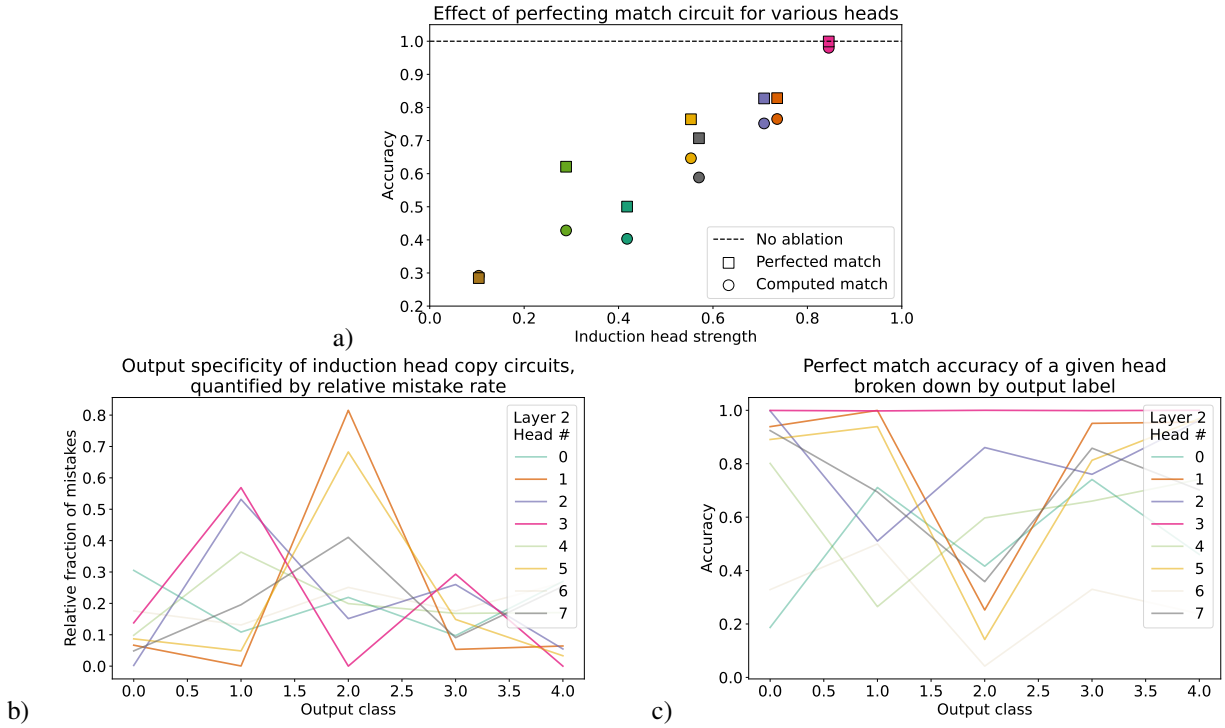

Figure 16. **a)** Induction heads are imperfect copiers. We consider the network from the end of training in Figure 3 and perform the ablate-all-but-head-X ablation used for the circles in Figure 4. We then use our artificial optogenetics framework to "perfect" each head's attention pattern (so it would have an induction strength of 1). While this increases the accuracy for most heads, the accuracy is still far from perfect, indicating that inducion heads at the end of training are imperfect copiers. **b)** We dive deeper and find label dependence in the imperfect copying. Specifically, here we plot the relative mistake rate by output label for each head. Line opacity is equal to induction head strength, to emphasize the more prominent induction heads. Interestingly, Head 3 is relatively worse at copying labels 1 and 3, but heads 1 and 5 compensate for this. Notably, in Figure 3, heads 1 and 5 appear after Head 3. We postulate that heads 1 and 5 are thus compensating for the mistakes that Head 3 makes. Future work could further explore such a "residual heads" hypothesis in more depth. **c)** For completeness, we include accuracy vs output class. We note that Head 3 with a "perfect" attention pattern overall does make very few mistakes, as we'd expect given its high accuracy in a). Notably, what plot b shows is that Head 3 with a "perfect" attention pattern, when it does make mistakes, tends to do so on classes 1 and 3.

that don't (rather than providing $4 \cdot 6 = 30$ new figures).

Figure 18 summarizes a reproduction of some of the results of Section 3.1. Specifically, strongest heads (determined via induction strength) are mostly able to solve the task. We find that the connection to the *neural race* (Saxe et al., 2022) mostly holds across seeds (first discussed in Section 3.2, though the correlation is not as strong as other metrics related to induction strength. Our main results on sub-circuit discovery (Figure 5) hold across seeds. The scaling result (Figure 6) reproduces across most seeds, though sometimes in standard training we see learning with 15 labels to be as quick as learning with 10 labels. For further details, we refer readers to the notebooks in our codebase.

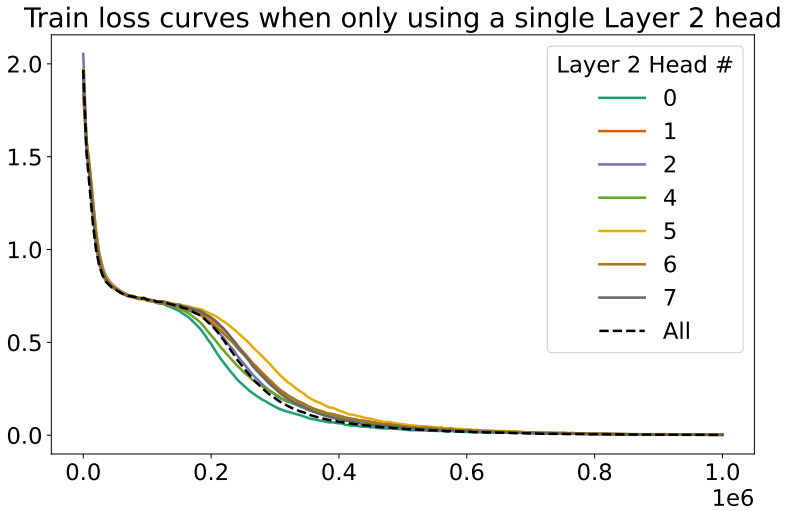

Figure 17. A follow-up to Figure 4b where we consider training with only a pair of Layer 2 heads active: Head 3 and one other head. Interestingly, loss curves snap to almost the true training loss curve. The relative ordering of phase changes is not indicative of the order of emergence (see Figure 3).

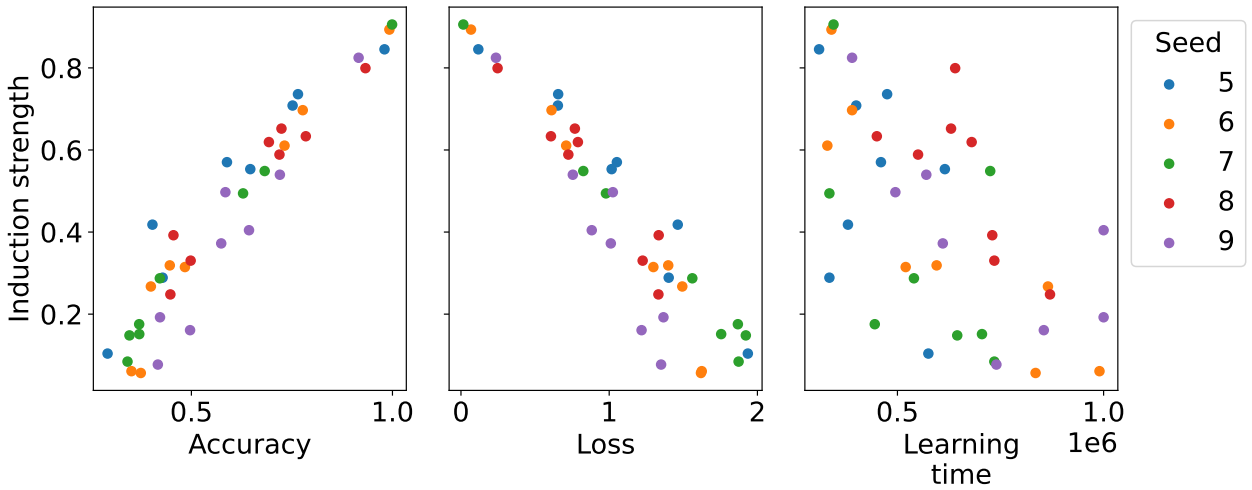

Figure 18. Reproducing some of the results of Section 3.1 across seeds. Specifically, we plot the induction strength of all 8 L2 heads across 5 seeds (40 points) versus various other metrics on individual heads: accuracy and loss if all other L2 heads are ablated (analogous to circles in Figure 4a,13), as well as the "learning time", which we define as the number of sequences seen such that the loss drops below $0.4 \cdot \log 2$ when training with a single head (a summary statistic for Figure 4b). As we can see, induction head strength is well correlated with accuracy and loss (lower loss is better, and corresponds to stronger heads) across seeds. We see that learning time (and thus, connections to the neural race reduction (Saxe et al., 2022)) is correlated to strength across seeds (faster learning times correlate to stronger heads), but not as strongly. Future work could investigate these ideas of circuits "racing" to minimize the loss further.